Point Distribution

Q1: Linear Algebra [20pts]

1. Inner Products [9pts]
2. Norms [6pts]
3. Eigen-stuff [5pts]

Q2: Working with Markov Chains [18pts Grad / 6% Bonus for Undergrads]

1. Markov Mechanics [6pts Grad / 2% Bonus for Undergrads]
2. Application to PageRank [12pts Grad / 4% Bonus for Undergrads]

Q3: Probability and Statistics [30pts]

1. Joint Distributions and Independence [6pts]
2. Covariance and Correlation [8pts]
3. Gaussian Distributions [6pts]
4. Conceptual Questions and Proofs [10pts]

Q4: Parameter Estimation Challenge [10pts]

Q5: Optimization [15pts + 6% Bonus for All]

1. KKT Conditions [15pts + 2% Bonus for All]
2. Using Jensen's for Convexity [4% Bonus for All]

Q6: Information Theory [35pts]

1. Entropy Proofs [20pts]
2. KL-Divergence [5pts]
3. Entropy and Informativity [10pts]

Q7: Entropy Bound [4% Bonus for All]

Points Totals:

- Non-Programming Base:
  - Undergrad: 110 pts
  - Graduate: 128 pts
- Non-Programming Bonus for Undergrad:
  - Undergrad: 6%
  - Graduate: these questions are required and included as base points
- Non-Programming Bonus for All:
  - Undergrad: 10%
  - Graduate: 10%
- For programming related points, see the Jupyter notebook.

1 Linear Algebra [20pts]

1.1 Inner Products [9pts]

In lecture, we discuss the dot product on $\mathbb{R}^n$. The dot product is an example of an **inner product**. Just like how there are different norms we can define, there are also different inner products that could be defined on $\mathbb{R}^n$. Furthermore, similar to norms, inner products (typically denoted as $\langle \cdot, \cdot \rangle$) have to satisfy a number of axioms.[1]

Just like how different norms give us different measures of length, different inner products give us different measures of *angles*. Across definitions of the inner product, orthogonality remains the same: $\vec{x}, \vec{y}$ are orthogonal if and only if $\langle \vec{x}, \vec{y} \rangle = 0$. However, since the inner product is a broad category of functions, it is very possible for two vectors that are orthogonal with respect to some inner product not to be orthogonal with respect to another inner product.

Consider the following vector.

$$\vec{v} = \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}$$

Given the following definitions of inner products, you need to find the *set of all vectors* orthogonal to that vector $\vec{v}$ under the specified definition of inner product. Show your work.

Hint

Your solution set may be infinite. If so, you should parametrize your result and state the range of the parameter.

a. $\langle (x_1, x_2, x_3), (y_1, y_2, y_3) \rangle \coloneqq x_1 y_1 + x_2 y_2 + x_3 y_3$. [3pts]
Substituting the values into the definition:

$$\langle \vec{v}, \vec{y} \rangle = 1(y_1) + 1(y_2) + 3(y_3) = y_1 + y_2 + 3y_3$$

Setting the inner product to 0 for orthogonality:

$$y_1 + y_2 + 3y_3 = 0 \implies y_1 = -y_2 - 3y_3$$

Parametrize the free variables y_2 and y_3. Let $y_2 = s$ and $y_3 = t$. Then:

$$\vec{y} = \begin{bmatrix} -s - 3t \\ s \\ t \end{bmatrix} = s \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix}$$

The set of orthogonal vectors is:

$$\left\{ s \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix} \mid s, t \in \mathbb{R} \right\}$$

b. $\langle (x_1, x_2, x_3), (y_1, y_2, y_3) \rangle \coloneqq 9x_1 y_1 + 4x_2 y_2 + x_3 y_3$. [3pts]
Substituting the values into the definition:

$$\langle \vec{v}, \vec{y} \rangle = 9(y_1) + 4(y_2) + 3(y_3) = 9y_1 + 4y_2 + 3y_3$$

Setting the inner product to 0 for orthogonality:

$$9y_1 + 4y_2 + 3y_3 = 0 \implies y_3 = -3y_1 - \frac{4}{3}y_2 = -3s - \frac{4}{3}t$$

The set of orthogonal vectors is:

$$\left\{ s \begin{bmatrix} 1 \\ 0 \\ -3 \end{bmatrix} + t \begin{bmatrix} 0 \\ 1 \\ -4/3 \end{bmatrix} \mid s, t \in \mathbb{R} \right\}$$

[1] For $\mathbb{R}^n$, generally, we look for Symmetry ($\forall \vec{x}, \vec{y};\ \langle \vec{x}, \vec{y} \rangle = \langle \vec{y}, \vec{x} \rangle$), Linearity in the First Argument ($\forall \vec{x}, \vec{y}, \vec{z}, \alpha \in \mathbb{R}, \beta \in \mathbb{R};\ \langle \alpha\vec{x} + \beta\vec{y}, \vec{z} \rangle = \alpha\langle \vec{x}, \vec{z} \rangle + \beta\langle \vec{y}, \vec{z} \rangle$), and Positive-Definiteness ($\forall \vec{x};\ \langle \vec{x}, \vec{x} \rangle = 0 \iff \vec{x} = \vec{0}$, and $\forall \vec{x};\ \langle \vec{x}, \vec{x} \rangle \geq 0$).

c. $\langle (x_1, x_2, x_3), (y_1, y_2, y_3) \rangle \coloneqq 6x_1y_1 - 2x_1y_2 - 2x_2y_1 + 3x_2y_2 + 3x_3y_3$. [3pts]
Substituting the values into the definition:

$$\langle \vec{v}, \vec{y} \rangle = 6(y_1) - 2(y_2) - 2(y_1) + 3(y_2) + 9(y_3) = 4y_1 + y_2 + 9y_3$$

Setting the inner product to 0 for orthogonality:

$$4y_1 + y_2 + 9y_3 = 0 \implies y_2 = -4y_1 - 9y_3$$

The set of orthogonal vectors is:

$$\left\{ s \begin{bmatrix} 1 \\ -4 \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ -9 \\ 1 \end{bmatrix} \mid s, t \in \mathbb{R} \right\}$$

1.2 Norms [6pts]

For each of the following functions, you must indicate whether they constitute a valid true norm. Recall from lecture, that a valid true norm satisfies positive-definiteness, absolute homogeneity, and the triangle inequality.

- If they *do not* constitute a valid true norm, you may merely indicate one property of norms they violate and provide an example input where the function violates that property.
- If they *do* constitute a valid true norm, you must prove that they satisfy all three properties of norms mentioned above.

Hint

The absolute value function $f(x) = |x|$ is a valid true norm on $\mathbb{R} \to \mathbb{R}_{\geq 0}$, so it will satisfy those properties. You may invoke this in your proofs as necessary.

a. [2pts] $f\colon \mathbb{R} \to \mathbb{R}_{\geq 0}$ with

$$f(x) = \left|\sin(x^2) + \frac{x+1}{\sqrt{3}} + 6x\right|.$$

This is not a valid norm. It violates Positive Definiteness. For a valid norm, $f(x) = 0 \iff x = 0$.

$$f(0) = \left|\sin(0^2) + \frac{0+1}{\sqrt{3}} + 6(0)\right| = \left|0 + \frac{1}{\sqrt{3}} + 0\right| = \frac{1}{\sqrt{3}} \neq 0$$

Since $f(0) \neq 0$, the function is not positive definite and therefore not a norm.

b. [2pts] $f\colon \mathbb{R}^2 \to \mathbb{R}_{\geq 0}$ with

$$f((x_1, x_2)) = \sqrt{|x_1|^2 + |x_2|^4}.$$

This is not a valid norm. It violates Absolute Homogeneity. For a valid norm, $f(\alpha\vec{x}) = |\alpha| f(\vec{x})$. Let $\vec{x} = (0, 1)$ and let scalar $\alpha = 2$.

$$|\alpha| f(\vec{x}) = |2| f((0,1)) = 2\sqrt{|0|^2 + |1|^4} = 2\sqrt{1} = 2$$

$$f(\alpha\vec{x}) = f(2(0,1)) = f((0,2)) = \sqrt{|0|^2 + |2|^4} = \sqrt{16} = 4$$

Since $4 \neq 2$, absolute homogeneity does not hold.

c. [2pts] $f\colon \mathbb{R}^3 \to \mathbb{R}_{\geq 0}$ with

$$f((x_1, x_2, x_3)) = |x_1| + 3|x_2| + 6|x_3|.$$

1. Positive Definiteness: Since $|x| \geq 0$ for all real x, $f(\vec{x}) \geq 0$. If $f(\vec{x}) = |x_1| + 3|x_2| + 6|x_3| = 0$, then since each term is non-negative, we must have $|x_1| = 0$, $3|x_2| = 0$, and $6|x_3| = 0$. This implies $x_1 = x_2 = x_3 = 0$, so $\vec{x} = \vec{0}$.

2. Absolute Homogeneity:

$$\begin{aligned} f(\alpha\vec{x}) &= |\alpha x_1| + 3|\alpha x_2| + 6|\alpha x_3| \\ &= |\alpha||x_1| + 3|\alpha||x_2| + 6|\alpha||x_3| \\ &= |\alpha|(|x_1| + 3|x_2| + 6|x_3|) \\ &= |\alpha| f(\vec{x}) \end{aligned}$$

3. Triangle Inequality: Let $\vec{x}, \vec{y} \in \mathbb{R}^3$. Using the triangle inequality for absolute values ($|a + b| \leq |a| + |b|$):

$$\begin{aligned} f(\vec{x} + \vec{y}) &= |x_1 + y_1| + 3|x_2 + y_2| + 6|x_3 + y_3| \\ &\leq (|x_1| + |y_1|) + 3(|x_2| + |y_2|) + 6(|x_3| + |y_3|) \\ &= (|x_1| + 3|x_2| + 6|x_3|) + (|y_1| + 3|y_2| + 6|y_3|) \\ &= f(\vec{x}) + f(\vec{y}) \end{aligned}$$

Since all properties are satisfied, f is a valid norm.

1.3 Eigen-stuff [5pts]

a. Let A be a real-valued, 2×2 matrix. You know that $\mathsf{det}(A) = 5$, and that $\mathsf{tr}(A) = 6$. What are the eigenvalues of A? Show your work. [2pts]

Let λ_1 and λ_2 be the eigenvalues of A.

$$\text{tr}(A) = \lambda_1 + \lambda_2 = 6$$

$$\det(A) = \lambda_1 \cdot \lambda_2 = 5$$

These eigenvalues are the roots of the characteristic polynomial $x^2 - \text{tr}(A)x + \det(A) = 0$:

$$x^2 - 6x + 5 = 0$$

$$(x-5)(x-1) = 0$$

Thus, the eigenvalues are $\lambda = 1$ and $\lambda = 5$.

b. Let B be a real-valued, symmetric, 3×3 matrix. $\vec{v}_1$ and $\vec{v}_2$ are eigenvectors of B that correspond to real, distinct eigenvalues λ_1, λ_2; they are as follows:

$$\vec{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \qquad \vec{v}_2 = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$$

Suppose that B has one more eigenvalue λ_3 that's different from the other 2. Find the third eigenvector, a non-zero vector $\vec{v}_3 \in \mathbb{R}^3$, such that $B\vec{v}_3 = \lambda_3\vec{v}_3$, and $\lambda_3 \neq \lambda_2, \lambda_3 \neq \lambda_1$. Show your work. [3pts]

Hint

This would be very difficult to deduce for a general matrix B, but we have that B is a real-valued symmetric matrix. What, then, can we conclude about its eigenvectors?

Since B is a real-valued symmetric matrix, the Spectral Theorem states that eigenvectors corresponding to distinct eigenvalues must be orthogonal.

Therefore, $\vec{v}_3$ must be orthogonal to both $\vec{v}_1 = [1, 2, 1]^\top$ and $\vec{v}_2 = [-2, 1, 0]^\top$.

$$\vec{v}_3 = \vec{v}_1 \times \vec{v}_2 = \begin{vmatrix} i & j & k \\ 1 & 2 & 1 \\ -2 & 1 & 0 \end{vmatrix}$$

$x_3 = (2)(0) - (1)(1) = -1$
$y_3 = -((1)(0) - (1)(-2)) = -2$
$z_3 = (1)(1) - (2)(-2) = 1 - (-4) = 5$

Thus, a valid third eigenvector is:

$$\vec{v}_3 = \begin{bmatrix} -1 \\ -2 \\ 5 \end{bmatrix}$$

2 Working with Markov Chains [18pts Grad / 6% Bonus for Undergrads]

2.1 Markov Mechanics [6pts Grad / 2% Bonus for Undergrads]

Suppose we have a sequence of random variables $\{X_t\}_{t\in\mathbb{N}}$ that take on values from the set $\mathcal{S} \coloneqq \{1, 2, \ldots, n\}$ (where n is just some positive integer.) We could think of the value of X_t as the state of some system at time t and we could think of $\mathcal{S}$ as the set of all possible states the system could be in. Naturally, we'd like to compute $P(X_t = x_t)$, the probability that the system is at state $x_t \in \mathcal{S}$ at time t. If we make the assumption that this is modelling a physical system, we can conclude the current state is dependent on the past, but not on the future. Thus, we can say:

$$P(X_t = x_t) = \sum_{x_{t-1}\in\mathcal{S}} \sum_{x_{t-2}\in\mathcal{S}} \cdots \sum_{x_0\in\mathcal{S}} P(X_t = x_t \mid X_{t-1} = x_{t-1}, \ldots, X_0 = x_0)P(X_{t-1} = x_{t-1}, \ldots, X_0 = x_0). \tag{1}$$

As it stands now, computing $P(X_t = x_t)$ for a fixed x_t would require us to add n^t terms and know all of the different n^t conditional probability distributions, which becomes intractible for relatively small values of n and t. To simplify, we can make an assumption known as the **first-order Markov assumption**:

$$P(X_t = x_t \mid X_0 = x_0, \ldots, X_{t-1} = x_{t-1}) = P(X_t = x_t \mid X_{t-1} = x_{t-1}),$$

for any $t \geq 1$. We've assumed that the current state is dependent only on the most recent state, and not the entire history. Our sequence $\{X_t\}_{t\in\mathbb{N}}$ is now called a **Markov chain**. You can verify that (1), under this assumption, becomes

$$P(X_t = x_t) = \sum_{x_{t-1}\in S} P(X_t = x_t \mid X_{t-1} = x_{t-1})P(X_{t-1} = x_{t-1}).$$

With this assumption, we only need to know the conditional probabilities for a single timestep, $P(X_t = i \mid X_{t-1} = j)$ for all $i, j \in \mathcal{S}$, and the distribution over the initial states, $P(X_0 = x_0)$ for any $x_0 \in \mathcal{S}$. These conditional probabilities are known as **transition probabilities**. It's also common to model these transition probabilities as constant over time, that is, $P(X_t = i \mid X_{t-1} = j) = P(X_q = i \mid X_{q-1} = j)$ for any $q, t \geq 1$ and $i, j \in \mathcal{S}$. To organize the transition probabilities, we can define a matrix $A \in \mathbb{R}^{n\times n}$ such that $A_{ij} \coloneqq P(X_t = i \mid X_{t-1} = j)$[2]. Because these are probabilities over all $\mathcal{S}$, all entries will be non-negative, and the columns will sum to 1:

$$\sum_{k=1}^{n} A_{kj} = 1 \quad \text{and} \quad A_{ij} \geq 0$$

Matrices with these properties are called **stochastic matrices**. If we let $\vec{x_0} \coloneqq \begin{bmatrix} P(X_0 = 1) & \ldots & P(X_0 = n) \end{bmatrix}^\top$, we can simulate a time-evolution of the system simply by multiplying by the matrix, $\vec{x_1} = A\vec{x_0}$[3]. More generally, the i-th entry of $\vec{x}_t \coloneqq A\vec{x}_{t-1}$ is $P(X_t = i)$. Crucially, since we assumed that the transition probabilities are constant over time, we don't even need a recurrence relation, we can just repeatedly multiply $\vec{x_0}$ by A, $\vec{x}_t = A^t\vec{x}_0$.

a. Given an $n\times n$ stochastic matrix A, <u>prove</u> that 1 is always an eigenvalue of A. [3pts Grad / 1% Bonus for Undergrads]

Hint

Recall that if λ is an eigenvalue of A, then $A\vec{v} = \lambda\vec{v} = (\lambda I)\vec{v}$, so $A\vec{v} - (\lambda I)\vec{v} = 0$, so $(A - \lambda I)\vec{v} = 0$, so $A - \lambda I$ must be singular. What corollaries of singularity might be easy to prove?

For $\lambda = 1$ to be an eigenvalue of A, we must show that $(A - I)$ is singular and has a determinant of 0. A matrix is singular if its rows or columns are linearly dependent. Looking at the sum of the entries in each column of the matrix $M = A - I$, the entries of M are given by $M_{ij} = A_{ij} - I_{ij}$. Summing the elements of the j-th column of M:

$$\sum_{i=1}^{n} M_{ij} = \sum_{i=1}^{n}(A_{ij} - I_{ij}) = \sum_{i=1}^{n} A_{ij} - \sum_{i=1}^{n} I_{ij}$$

We are given that A is a stochastic matrix, meaning the sum of each column is 1. Also, since I is the identity matrix, each column contains exactly one 1 and zeros elsewhere, so $\sum_{i=1}^{n} I_{ij} = 1$.

[2] You'll have to use 1-indexing if you want this to align with our definition $\mathcal{S} \coloneqq \{1, 2, \ldots, n\}$.

[3] It may be instructive to work out a single element of $\vec{x_1}$ symbolically, and confirm that it does correctly evaluate the sum rule using the transition probabilities to get the probability of the system at the next timestep.

Substituting these values back:

$$\sum_{i=1}^{n} M_{ij} = 1 - 1 = 0$$

Since the sum of the elements in every column of $(A - I)$ is 0, the rows of $(A - I)$ are linearly dependent. Linearly dependent rows imply that the matrix is singular, and therefore $\det(A - I) = 0$. Thus, 1 is an eigenvalue of A.

b. Let's interpret that result. Given an eigenvalue λ of A, we know that there must exist an eigenvector $\vec{v}$ such that $A\vec{v} = \lambda\vec{v}$. With $\lambda = 1$, we know further that $A\vec{v} = \vec{v}$. Interpreting A as a transformation, $\vec{v}$ is unaffected by this transformation. When such a $\vec{v}$ is a *probability vector* (i.e., a vector whose entries are non-negative and sum to 1), then interpreting $\vec{v}$ as a distribution over our states $\mathcal{S}$ means that this distribution over the states is unaffected by time-evolution. Thus, it's called a *steady-state vector*. When the state of our system has approached a steady-state vector with $A\vec{v} = \vec{v}$, then our belief of the state of the system doesn't change.

Given the stochastic matrix

$$A = \begin{bmatrix} 1/2 & 2/3 \\ 1/2 & 1/3 \end{bmatrix},$$

find all steady-state vectors of the corresponding Markov chain. Remember to normalize your result to a probability vector. <u>Show your work.</u> [3pts Grad / 1% Bonus for Undergrads]

A steady-state vector $\vec{v}$ satisfies the equation $A\vec{v} = \vec{v}$, which is equivalent to $(A - I)\vec{v} = \vec{0}$.

First, compute $A - I$:

$$A - I = \begin{bmatrix} 1/2 & 2/3 \\ 1/2 & 1/3 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -1/2 & 2/3 \\ 1/2 & -2/3 \end{bmatrix}$$

Solve the homogeneous system $(A - I)\vec{v} = \vec{0}$ for $\vec{v} = [v_1, v_2]^\top$:

$$\begin{bmatrix} -1/2 & 2/3 \\ 1/2 & -2/3 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

The two rows are linearly dependent so solve the first row equation:

$$-\frac{1}{2}v_1 + \frac{2}{3}v_2 = 0 \implies \frac{1}{2}v_1 = \frac{2}{3}v_2 \implies 3v_1 = 4v_2$$

Let $v_2 = 3t$. Then $3v_1 = 4(3t) \implies v_1 = 4t$. Thus, the eigenvectors are of the form:

$$\vec{v} = t \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

To be a valid probability vector, the entries must sum to 1:

$$v_1 + v_2 = 1 \implies 4t + 3t = 1 \implies 7t = 1 \implies t = \frac{1}{7}$$

Substituting $t = 1/7$ back into our vector:

$$\vec{v} = \frac{1}{7} \begin{bmatrix} 4 \\ 3 \end{bmatrix} = \begin{bmatrix} 4/7 \\ 3/7 \end{bmatrix}$$

The unique steady-state vector is:

$$\vec{v} = \begin{bmatrix} 4/7 \\ 3/7 \end{bmatrix}$$

2.2 Application to PageRank [12pts Grad / 4% Bonus for Undergrads]

The Google PageRank algorithm is a classic application of Markov chains. The main objective is ranking webpages by their importance, which we roughly take to mean the proportion of time a user spends on a webpage as they surf the Web. Each webpage is a state, and the transition probabilities dictate how a user moves between the webpages that are hyperlinked on the current webpage. To satisfy some regularity conditions, slight adjustments are made to the stochastic matrix A (i.e., the final stochastic matrix we get isn't going to only be made up of the transition probabilities as was explained in the previous part):

- If there exists a column of A with only a single entry being 1, replace the whole column of A with $1/n$, where n is the number of webpages in total, to get a new matrix A'.
- We then compute the Google matrix G, where

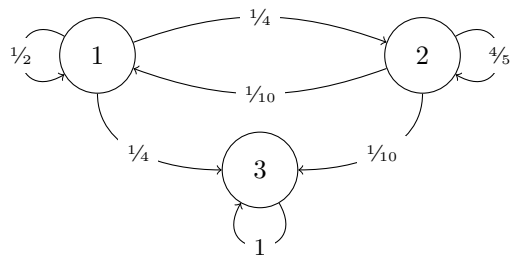
$$G \coloneqq \alpha A' + (1-\alpha)K,$$

  $0 < \alpha < 1$, and K is the matrix with all entries equal to $1/n$. The idea is that sometimes a websurfer will close the tab and start a new search at a random site.
- G is the stochastic matrix that represents an appropriate Markov chain for modelling our problem.

Now, G must have a unique steady state $\vec{x}^*$, that is, for any valid initial distribution $\vec{x}_0$, we have that

$$\vec{x}^* \coloneqq \lim_{k\to\infty} \vec{x}_k = \lim_{k\to\infty} G^k \vec{x}_0 \implies G\vec{x}^* = \vec{x}^*$$

Suppose we have 3 webpages on the Web. The following graph represents the different transition probabilities between the webpages.



For the sake of cleanliness, we've excluded any edges with a weight of 0, but all other edges should be present and interpretable as transition probabilities. For example, we see here that $P(X_t = 2 \mid X_{t-1} = 1) = \frac{1}{4}$.
You'll be applying PageRank to this tiny web. From user surveying, we've found that $\alpha = 0.8$ is a good choice for our modelling purposes.

a. Construct the base stochastic matrix, A. [1.5pts Grad / 0.5% Bonus for Undergrads]

$$A = \begin{bmatrix} 1/2 & 1/10 & 0 \\ 1/4 & 4/5 & 0 \\ 1/4 & 1/10 & 1 \end{bmatrix}$$

b. Construct G. <u>Show your work.</u> [6pts Grad / 2% Bonus for Undergrads]
Column 3 has a single entry of 1, so replace it with $[1/3, 1/3, 1/3]^\top$ to get A':

$$A' = \begin{bmatrix} 1/2 & 1/10 & 1/3 \\ 1/4 & 4/5 & 1/3 \\ 1/4 & 1/10 & 1/3 \end{bmatrix}$$

Calculate $G = \alpha A' + (1-\alpha)K$ with $\alpha = 0.8 = 4/5$ and $K_{ij} = 1/3$.

$$G = \frac{4}{5}A' + \frac{1}{5}K = \frac{4}{5}\begin{bmatrix} 1/2 & 1/10 & 1/3 \\ 1/4 & 4/5 & 1/3 \\ 1/4 & 1/10 & 1/3 \end{bmatrix} + \begin{bmatrix} 1/15 & 1/15 & 1/15 \\ 1/15 & 1/15 & 1/15 \\ 1/15 & 1/15 & 1/15 \end{bmatrix}$$

Column 1:

$$\frac{4}{5}[1/2, 1/4, 1/4]^\top + 1/15 = [2/5, 1/5, 1/5]^\top + 1/15 = [7/15, 4/15, 4/15]^\top = [35/75, 20/75, 20/75]^\top$$

Column 2:

$$\frac{4}{5}[1/10, 4/5, 1/10]^\top + 1/15 = [2/25, 16/25, 2/25]^\top + 1/15 = [11/75, 53/75, 11/75]^\top$$

Column 3:

$$\frac{4}{5}[1/3, 1/3, 1/3]^\top + 1/15 = [4/15, 4/15, 4/15]^\top + 1/15 = [5/15, 5/15, 5/15]^\top = [25/75, 25/75, 25/75]^\top$$

$$G = \begin{bmatrix} 35/75 & 11/75 & 25/75 \\ 20/75 & 53/75 & 25/75 \\ 20/75 & 11/75 & 25/75 \end{bmatrix}$$

c. Find the unique steady-state vector of G. <u>Show your work.</u> [3pts Grad / 1% Bonus for Undergrads]
Solve $(G - I)\vec{x} = 0$. Multiplying by 75 to clear denominators:

$$\begin{bmatrix} 35-75 & 11 & 25 \\ 20 & 53-75 & 25 \\ 20 & 11 & 25-75 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -40 & 11 & 25 \\ 20 & -22 & 25 \\ 20 & 11 & -50 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \vec{0}$$

From row 3 - row 2:

$$(20-20)x_1 + (11-(-22))x_2 + (-50-25)x_3 = 0 \implies 33x_2 = 75x_3 \implies 11x_2 = 25x_3 \implies x_2 = \frac{25}{11}x_3$$

Substitute into row 3 equation ($20x_1 + 11x_2 - 50x_3 = 0$):

$$20x_1 + 11(\frac{25}{11}x_3) - 50x_3 = 0 \implies 20x_1 + 25x_3 - 50x_3 = 0 \implies 20x_1 = 25x_3 \implies x_1 = \frac{5}{4}x_3$$

Let $x_3 = 44$. Then $x_1 = 55$ and $x_2 = 100$. The sum is $55 + 100 + 44 = 199$.

$$\vec{x}^* = \begin{bmatrix} 55/199 \\ 100/199 \\ 44/199 \end{bmatrix}$$

d. Order the webpages by how frequently a user would visit the webpages as they surf the Web indefinitely. [1.5pts Grad / 0.5% Bonus for Undergrads]
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Hint

Your work will be easier to evaluate if you do not round intermediate results. You should represent all numbers appearing in your work as fractions or integers. When simplified, the entries for the steady-state vector should have 199 as a denominator.

3 Probability and Statistics [30pts]

3.1 Joint Distributions and Independence [6pts]

The following table summarizes data collected from 10 students. Let X be a random variable that is 1 if the student studied for an exam and 0 otherwise. Let Y be a random variable that is 1 if the student passed the exam and 0 otherwise.

	$Y = 1$ (Passed)	$Y = 0$ (Failed)	Total
$X = 1$ (Studied)	4	2	6
$X = 0$ (Didn't Study)	1	3	4
Total	5	5	10

Answer the following questions using the empirical data about X and Y in the table above.

a. Compute $P(X = 1, Y = 1)$. [1pts] 0.4

b. Compute the marginal probabilities $P(X = 1)$ and $P(Y = 1)$. [1.5pts]

$$P(X = 1) = 0.6, P(Y = 1) = 0.5$$

c. Compute the conditional probability $P(Y = 1 \mid X = 1)$. [1.5pts]

$$P(Y = 1 \mid X = 1) = \frac{P(X = 1, Y = 1)}{P(X = 1)} = \frac{0.4}{0.6} = \frac{2}{3}$$

d. Are X and Y statistically independent? Justify your answer using your previous calculations. [2pts]
Two random variables are statistically independent if and only if $P(X, Y) = P(X)P(Y)$.

$$P(X = 1, Y = 1) = 0.4$$

$$P(X = 1) \cdot P(Y = 1) = 0.6 \cdot 0.5 = 0.3$$

Since $0.4 \neq 0.3$, X and Y are not statistically independent.

3.2 Covariance and Correlation [8pts]

a. Suppose X and Y are two random variables. X has the pmf:

$$p(x) = \begin{cases} 0.3 & x = c \\ 0.7 & x = -c \end{cases}$$

where c is a nonzero constant. Let Y obey the Standard Normal distribution, $Y \sim \mathcal{N}(0, 1)$.

X and Y are statistically independent. Define a new random variable $Z = X - Y$. Calculate the covariance of Y and Z, i.e., $\mathsf{Cov}(Y, Z)$. Show your work. [5pts]
Using the linearity property of covariance:

$$\text{Cov}(Y, Z) = \text{Cov}(Y, X - Y) = \text{Cov}(Y, X) - \text{Cov}(Y, Y)$$

Given that X and Y are statistically independent, which implies that their covariance is zero:

$$\text{Cov}(Y, X) = 0$$

The covariance of a variable with itself is its variance:

$$\text{Cov}(Y, Y) = \text{Var}(Y)$$

Since $Y \sim \mathcal{N}(0, 1)$, its variance is 1:

$$\text{Var}(Y) = 1$$

Substituting these values back into our equation:

$$\text{Cov}(Y, Z) = 0 - 1 = -1$$

b. Let A and B be statistically independent random variables with $\mathsf{Var}(A) = 2$ and $\mathsf{Var}(B) = 15$. Define a new random variable $C = 4A + 2B$. The correlation coefficient, $\rho(A, C)$, is defined as:

$$\rho(A, C) = \frac{\mathsf{Cov}(A, C)}{\sqrt{\mathsf{Var}(A)\,\mathsf{Var}(C)}}$$

Calculate $\rho(A, C)$, the correlation coefficient between A and C. Show your work. [3pts]
Given $C = 4A + 2B$, $\text{Var}(A) = 2$, $\text{Var}(B) = 15$, and A and B are independent, $\text{Cov}(A, B) = 0$.

$$\begin{aligned} \text{Cov}(A, C) &= \text{Cov}(A, 4A + 2B) \\ &= 4\text{Cov}(A, A) + 2\text{Cov}(A, B) \\ &= 4\text{Var}(A) + 2(0) \\ &= 4(2) = 8 \end{aligned}$$

$$\begin{aligned} \text{Var}(C) &= \text{Var}(4A + 2B) \\ &= 4^2\text{Var}(A) + 2^2\text{Var}(B) \\ &= 16(2) + 4(15) \\ &= 32 + 60 = 92 \end{aligned}$$

$$\rho(A, C) = \frac{\text{Cov}(A, C)}{\sqrt{\text{Var}(A)\text{Var}(C)}} = \frac{8}{\sqrt{2 \cdot 92}} = \frac{8}{\sqrt{184}} = \frac{4}{\sqrt{46}}$$

3.3 Gaussian Distributions vs. Gaussian-Distributed Random Variables [6pts]

You are given two **independent** random variables, $X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2)$ and $X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$. These are "*Gaussian-distributed random variables*," that is, their probability density functions are described by the Gaussian function:

$$\mathcal{N}(\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \cdot \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

a. Let $Y = X_1 + X_2$. One way to conceptualize sampling a value for Y is to imagine sampling a value for both X_1 and X_2, then adding the results of those trials. Curiously, the probability density function of Y is also a Gaussian. The standard result for $Y = X_1 + X_2$ would be $Y \sim \mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$. Proving that Y is actually Gaussian-distributed requires some tools we haven't taught, but a much easier result is to confirm the mean and variance.

   Show that $\mathbb{E}[Y] = \mu_1 + \mu_2$ and $\mathsf{Var}(Y) = \sigma_1^2 + \sigma_2^2$. <u>Show your work.</u> [3pts]
   Using the linearity of expectation:

$$\mathbb{E}[Y] = \mathbb{E}[X_1 + X_2] = \mathbb{E}[X_1] + \mathbb{E}[X_2]$$

   Since $X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2)$ and $X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$, $\mathbb{E}[X_1] = \mu_1$ and $\mathbb{E}[X_2] = \mu_2$.

$$\mathbb{E}[Y] = \mu_1 + \mu_2$$

   variance of a sum of random variables is given by:

$$\text{Var}(Y) = \text{Var}(X_1 + X_2) = \text{Var}(X_1) + \text{Var}(X_2) + 2\text{Cov}(X_1, X_2)$$

   Since X_1 and X_2 are independent, covariance is zero ($\text{Cov}(X_1, X_2) = 0$). Thus:

$$\text{Var}(Y) = \text{Var}(X_1) + \text{Var}(X_2) = \sigma_1^2 + \sigma_2^2$$

b. Now consider a new random variable Z, defined as:

$$Z = \begin{cases} X_1 & \text{with probability } 0.5 \\ X_2 & \text{with probability } 0.5 \end{cases}$$

   Can Z be described as a Gaussian random variable? <u>Briefly (1-2 sentences or quick math) justify</u> your choice. [3pts]
   Z is not a Gaussian random variable. Z is a mixture of two Gaussians, meaning its PDF is the weighted average of the two Gaussian PDFs: $P(Z) = 0.5P(X_1) + 0.5P(X_2)$. This distribution is multimodal, whereas a single Gaussian distribution is unimodal.

3.4 Conceptual Questions and Proofs [10pts]

a. Variance is defined as the expected amount by which data varies from the mean, μ.

$$\mathsf{Var}(X) = \mathbb{E}[(X-\mu)^2]$$

You can also formulate variance using only the first- and second-order moments of the random variable.

$$\mathsf{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$

Using expectation identities, <u>prove</u> that these two formulations are equivalent. [5pts]

$$\mathrm{Var}(X) = \mathbb{E}[(X-\mu)^2]$$
$$(X-\mu)^2 = X^2 - 2X\mu + \mu^2$$

By the linearity of expectation, $\mathbb{E}[A+B] = \mathbb{E}[A] + \mathbb{E}[B]$ and $\mathbb{E}[cX] = c\mathbb{E}[X]$ for any constant c.

$$\begin{aligned}\mathrm{Var}(X) &= \mathbb{E}[X^2 - 2X\mu + \mu^2] \\ &= \mathbb{E}[X^2] - \mathbb{E}[2X\mu] + \mathbb{E}[\mu^2] = \mathbb{E}[X^2] - 2\mu\mathbb{E}[X] + \mu^2\end{aligned}$$

Substitute $\mathbb{E}[X]$ back in for μ:

$$\begin{aligned}&= \mathbb{E}[X^2] - 2(\mathbb{E}[X])(\mathbb{E}[X]) + (\mathbb{E}[X])^2 \\ &= \mathbb{E}[X^2] - 2(\mathbb{E}[X])^2 + (\mathbb{E}[X])^2 \\ &= \mathbb{E}[X^2] - (\mathbb{E}[X])^2\end{aligned}$$

Thus, $\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$.

b. <u>Explain</u> in your own words why two random variables can be uncorrelated (i.e., $\mathsf{Cov}(A,B) = 0$) but also not statistically independent (i.e., $P(A,B) \neq P(A) \cdot P(B)$). Also, <u>provide one simple example</u> of two random variables that demonstrate this property. [5pts]

Correlation measures the strength of the linear relationship between two variables. Independence implies that knowing the value of one variable gives no information about the other. Two variables can have a strong relationship that is non-linear. In such a case, they are dependent, but if the relationship is symmetric, the linear trends may cancel out, resulting in zero correlation.

Example:

Let X be a random variable uniformly distributed on the set $\{-1, 0, 1\}$. Let $Y = X^2$. X and Y are not independent because if you know $X = 0$, you know that $Y = 0$.

However, they are uncorrelated:

- $\mathbb{E}[X] = \frac{-1+0+1}{3} = 0$.
- $\mathbb{E}[XY] = \mathbb{E}[X^3]$. The values of X^3 are $(-1)^3 = -1$, $0^3 = 0$, $1^3 = 1$.
- $\mathbb{E}[XY] = \frac{-1+0+1}{3} = 0$.
- $\mathrm{Cov}(X,Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = 0 - (0)\mathbb{E}[Y] = 0$.

Since the covariance is 0, they are uncorrelated, despite being perfectly dependent.

4 Parameter Estimation Challenge [10pts]

You're about to play a game of chance, and your opponent provides a coin. They say, "don't worry, it's a fair coin," but because they even thought to say that, now you don't trust them.

We can model the coin as a random variable C with a single parameter $\theta \in [0, 1]$, the chance of landing on heads.

$$C = \begin{cases} \text{Heads} & p = \theta \\ \text{Tails} & p = 1 - \theta \end{cases}$$

You flip the coin 10 times and get 7 heads and 3 tails (these tosses are independent). This isn't promising, but not disqualifying, given the small sample size. While flipping the coin, you see it tumble and feel its weight, and you don't get the sense that the coin is biased. We can model your belief about θ with the following distribution:

$$P(\theta) \propto \theta^5 (1 - \theta)^5$$

It has a mode at 0.5, but it's not so tight, modelling a healthy distrust of your opponent.

In class, you will have learned about Maximum Likelihood Estimation (MLE). You recorded some data—7 heads and 3 tails—and you could pick a θ that maximizes the likelihood of that data. In MLE, we simply pick:

$$\theta^* := \arg\max_{\theta} P(D|\theta)$$

However, this doesn't take into account any prior belief. We need a way to include your prior probability distribution. Bayes Rule gives us a way to update likelihoods with prior beliefs. We can instead do *Maximum A Posteriori* (MAP) estimation:

$$\theta^* := \arg\max_{\theta} P(\theta|D) = \arg\max_{\theta} \frac{P(D|\theta) \cdot P(\theta)}{P(D)} = \arg\max_{\theta} P(D|\theta) \cdot P(\theta)$$

Since $P(D)$ is constant w.r.t. θ, we can simply remove it from the argmax, since it would scale every choice by the same factor. Thus we're left with $P(D|\theta)$, the likelihood of our observed data, and $P(\theta)$, our prior belief about the coin.

MLE would give you $\theta^* = 0.7$ as the best estimator for the coin's parameter. What value of θ is the best estimator for the coin's parameter using your prior and MAP estimation? You should first find a statement for the posterior probability $P(\theta|D)$ up to any proportionality constants, then maximize that expression. Show your work. [10pts]

Hint

The expression you find may not be easy to optimize. One trick very useful in optimization is composition. If you compose a function with any monotonically increasing function, then the extrema will remain extrema, and their positions will remain the same. Thus, if you're trying to optimize $f(x)$ on [0,1], you can find any function that's increasing on [0,1] and compose f with it.

$$\arg\max_{x \in [0,1]} f(x) = \arg\max_{x \in [0,1]} \ln(f(x)) = \arg\max_{x \in [0,1]} (f(x))^2 = \arg\max_{x \in [0,1]} e^{f(x)} = \dots$$

To find the MAP, maximize the posterior probability $P(\theta|D)$ and the posterior is proportional to the likelihood times the prior (Bayes' Rule)

$$P(\theta|D) \propto P(D|\theta) \cdot P(\theta)$$

Since the coin flips are independent, the likelihood of 7 heads and 3 tails has Binomial distribution:

$$P(D|\theta) \propto \theta^7 (1 - \theta)^3$$

$$P(\theta|D) \propto \left(\theta^7 (1 - \theta)^3\right) \cdot \left(\theta^5 (1 - \theta)^5\right)$$

$$P(\theta|D) \propto \theta^{12} (1 - \theta)^8$$

To find θ^*, maximize the posterior. Specifically, it is easier to maximize the log posterior, $L(\theta) = \ln(P(\theta|D))$:

$$L(\theta) = \ln(\theta^{12}(1 - \theta)^8) = 12 \ln(\theta) + 8 \ln(1 - \theta)$$

$$\frac{dL}{d\theta} = \frac{12}{\theta} - \frac{8}{1 - \theta}$$

Setting the derivative to 0 to find the max:

$$\frac{12}{\theta} = \frac{8}{1-\theta}$$

$$12(1-\theta) = 8\theta \implies 12 - 12\theta = 8\theta \implies 20\theta = 12$$

$$\theta = \frac{12}{20} = 0.6$$

Thus, the MAP estimate for the coin's parameter is 0.6.

5 Optimization [15pts + 6% Bonus for All]

5.1 KKT Conditions [15pts + 2% Bonus for All]

Consider the following minimization problem:

$$\begin{aligned}
\min_{\vec{x}\in\mathbb{R}^3} \quad & f(\vec{x}) = 3x_1^2 + 5x_2^2 - x_2x_3 + 2x_3^2, \qquad \vec{x} = (x_1, x_2, x_3) \\
\text{s.t.} \quad & g_1(\vec{x}) = x_1 + x_2 + x_3 \leq -4 \\
& g_2(\vec{x}) = 4x_1^2 \leq 5 \\
& g_3(\vec{x}) = 3x_2 + 8x_1 \leq -50 \\
& g_4(\vec{x}) = -x_3^2 \leq -6
\end{aligned}$$

a. Write the Lagrange function, $\mathcal{L}(\vec{x}; \lambda_1, \lambda_2, \lambda_3, \lambda_4)$ for this minimization problem. [3pts]

$$\mathcal{L}(\vec{x}, \vec{\lambda}) = (3x_1^2 + 5x_2^2 - x_2x_3 + 2x_3^2) + \lambda_1(x_1 + x_2 + x_3 + 4) + \lambda_2(4x_1^2 - 5) + \lambda_3(8x_1 + 3x_2 + 50) + \lambda_4(-x_3^2 + 6)$$

b. List the names of all 4 groups of KKT conditions and their corresponding mathematical equations or inequalities for this specific minimization problem. Show your work. [9pts]

1. Stationarity ($\nabla_{\vec{x}}\mathcal{L} = 0$):

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial x_1} &= 6x_1 + \lambda_1 + 8x_1\lambda_2 + 8\lambda_3 = 0 \\
\frac{\partial \mathcal{L}}{\partial x_2} &= 10x_2 - x_3 + \lambda_1 + 3\lambda_3 = 0 \\
\frac{\partial \mathcal{L}}{\partial x_3} &= -x_2 + 4x_3 + \lambda_1 - 2x_3\lambda_4 = 0
\end{aligned}$$

2. Primal Feasibility ($g_i(\vec{x}) \leq 0$):

$$\begin{aligned}
x_1 + x_2 + x_3 + 4 &\leq 0 \\
4x_1^2 - 5 &\leq 0 \\
8x_1 + 3x_2 + 50 &\leq 0 \\
-x_3^2 + 6 &\leq 0
\end{aligned}$$

3. Dual Feasibility ($\lambda_i \geq 0$):

$$\lambda_1 \geq 0, \quad \lambda_2 \geq 0, \quad \lambda_3 \geq 0, \quad \lambda_4 \geq 0$$

4. Complementary Slackness ($\lambda_i g_i(\vec{x}) = 0$):

$$\begin{aligned}
\lambda_1(x_1 + x_2 + x_3 + 4) &= 0 \\
\lambda_2(4x_1^2 - 5) &= 0 \\
\lambda_3(8x_1 + 3x_2 + 50) &= 0 \\
\lambda_4(-x_3^2 + 6) &= 0
\end{aligned}$$

c. In addition to the Lagrange multipliers, KKT introduces the idea of constraint activity. When enumerating all possible constraint activity sets, how many systems would we need to solve? [3pts]
16 systems

d. Find the constrained global minimum! This is quite a lot of work and we don't expect you to do it by hand; you may use any tool you like. Once you set activity, KKT reduces to a system of equalities (and you check the solutions for feasibility after solving), so any tool for solving equations would work.

There are several capable python packages, including SymPy or sagemath. There may be online tools, but those tend to be quite limited, especially in the number of variables allowed.

List out all feasible candidates (including their Lagrange multipliers), then perform the candidates test. Round final values to 6 decimal points. [2% Bonus for All]

The constraint $g_4(\vec{x}) = -x_3^2 + 6 \leq 0$ implies $|x_3| \geq \sqrt{6}$. This splits region into positive ($x_3 \geq 2.45$) and negative ($x_3 \leq -2.45$) components. The objective function contains the term $-x_2x_3$. Since g_3 forces x_2 to be negative (≈ -13.7), minimizing f requires x_3 to be negative (so the product $-x_2x_3$ becomes a large subtracting term). Optimization tools often get stuck in the local minimum ($x_3 \approx 2.45$, $f \approx 985$) if initialized with a positive x_3. The true global minimum lies in the negative region.

Solving the KKT system for active g_2, g_3 and inactive g_4:

$$
\begin{aligned}
x_1 &= -\frac{\sqrt{5}}{2} \approx -1.118034 \\
x_2 &= \frac{-50 + 4\sqrt{5}}{3} \approx -13.685249 \\
x_3 &= \frac{x_2}{4} \approx -3.421312
\end{aligned}
$$

Since $|x_3| \approx 3.42 > \sqrt{6}$, the constraint g_4 is satisfied, confirming $\lambda_4 = 0$.

- $\mathbf{x_1} = -1.118034$
- $\mathbf{x_2} = -13.685249$
- $\mathbf{x_3} = -3.421312$
- $\lambda_\mathbf{1} = 0$
- $\lambda_\mathbf{2} \approx 39.031498$
- $\lambda_\mathbf{3} \approx 44.477060$
- $\lambda_\mathbf{4} = 0$

Function Value: $f(\vec{x}^*) \approx 916.037169$

Example: Using Jensen's Inequality for Gibbs' Inequality [0pts]

Jensen's inequality is a very powerful tool in ML. In general,

$$\begin{cases} \mathbb{E}\left[f(X)\right] \geq f\left(\mathbb{E}[X]\right) & \text{when f is convex} \\ \mathbb{E}\left[f(X)\right] \leq f\left(\mathbb{E}[X]\right) & \text{when f is concave} \end{cases}$$

Choosing the right composition function f can generate very useful results. For example, one can use Jensen's inequality to prove Gibbs' inequality, that is, that KL-Divergence is non-negative. For discrete probability distributions P and Q,

$$D_{\mathrm{KL}}(P\|Q) = \sum_i p_i \log \frac{p_i}{q_i}$$

We can express this as an expectation of $\log \frac{p_i}{q_i}$ over the weights p_i. For Jensen's we also need a convex function to get $D_{\mathrm{KL}} \geq \cdots$ and not $D_{\mathrm{KL}} \leq \cdots$; log is concave, but then $-\log$ is convex, and that's just a log rule away.

$$D_{\mathrm{KL}}(P\|Q) = \sum_i p_i \cdot \left(\log \frac{p_i}{q_i}\right) = \mathbb{E}_p\left[\log \frac{p_i}{q_i}\right] = \mathbb{E}_p\left[-\log \frac{q_i}{p_i}\right]$$

To appropriately apply Jensen's, we need to show that $-\log$ is convex. Briefly, $f(x) = -\log(x)$, $f'(x) = \frac{-1}{\ln(2)x}$, $f''(x) = \frac{1}{\ln(2)x^2}$. $x^2 \geq 0$, so $\frac{1}{x^2} \geq 0$, so $\frac{1}{\ln(2)x^2} \geq 0$. Thus it's convex, so:

$$\mathbb{E}_p\left[-\log \frac{q_i}{p_i}\right] \geq -\log\left(\mathbb{E}_p\left[\frac{q_i}{p_i}\right]\right)$$

$$D_{\mathrm{KL}}(P\|Q) = \mathbb{E}_p\left[-\log \frac{q_i}{p_i}\right] \geq -\log\left(\mathbb{E}_p\left[\frac{q_i}{p_i}\right]\right) = -\log\left(\sum_i p_i \cdot \frac{q_i}{p_i}\right) = -\log\left(\sum_i q_i\right) = -\log(1) = 0$$

$$D_{\mathrm{KL}}(P\|Q) \geq 0$$

Thus, we've shown it directly. □

In other situations, this may not be how you utilize Jensen's. The formulation of KL-divergence lends itself nicely to direct application of Jensen's, but sometimes you may need to work backward, taking what you want to find and constructing an arbitrary set of inputs that will result in your target.

5.2 Using Jensen's for Convexity [4% Bonus for All]

For a multivariate function $g : \mathbb{R}^n \to \mathbb{R}$, you can check its convexity by finding it's Hessian matrix. If the Hessian is positive semi-definite, then the function is convex. The Hessian is defined as:

$$\boldsymbol{H_g} = \nabla^2 g(x) = \begin{bmatrix} \frac{\partial^2 g}{\partial x_1^2} & \frac{\partial^2 g}{\partial x_1 \partial x_2} & \cdots \\ \frac{\partial^2 g}{\partial x_2 \partial x_1} & \frac{\partial^2 g}{\partial x_2^2} & \\ \vdots & & \ddots \end{bmatrix}$$

Next, we can say a matrix is positive semi-definite if its quadratic form w.r.t. an **arbitrary** vector $\vec{d} \in \mathbb{R}^n$ is always non-negative:

$$\vec{d}^\top \boldsymbol{H_g} \vec{d} \overset{?}{\geq} 0$$

Consider the log-sum-exp function $f(x) = \ln(\sum_{i=1}^n e^{x_i})$. Using this process, <u>prove</u> that f is convex on $\mathbb{R}^n$, that is, that $\vec{d}^\top \boldsymbol{H_f} \vec{d} \geq 0$. There are many ways to prove this, but for this question, **<u>you must use Jensen's inequality</u>** as the crux of your proof.

Hint

Let $\phi(t) = t^2$. $\phi''(t) = 2 \geq 0$, so ϕ is convex. One possible solution path will use ϕ as the composition function for Jensen's.

1. Calculate the Hessian Matrix
find the first partial derivatives. Let $S = \sum_{k=1}^{n} e^{x_k}$.

$$\frac{\partial f}{\partial x_i} = \frac{e^{x_i}}{S}$$

Let $p_i = \frac{e^{x_i}}{S}$. Observe that $p_i > 0$ and $\sum p_i = 1$, so p represents a discrete probability distribution. Compute the second partial derivatives to form the Hessian H:

- For diagonal elements ($i = j$): $\frac{\partial^2 f}{\partial x_i^2} = \frac{e^{x_i} S - (e^{x_i})^2}{S^2} = p_i - p_i^2$.
- For off-diagonal elements ($i \neq j$): $\frac{\partial^2 f}{\partial x_i \partial x_j} = \frac{-e^{x_i} e^{x_j}}{S^2} = -p_i p_j$.

Thus, the Hessian can be written in matrix form as $H = \text{diag}(p) - \vec{p}\vec{p}^\top$.
2. Quadratic Form Analysis
To prove convexity, show $\vec{d}^\top H \vec{d} \geq 0$ for any vector $\vec{d} \in \mathbb{R}^n$.

$$\begin{aligned}
\vec{d}^\top H \vec{d} &= \vec{d}^\top (\text{diag}(p) - \vec{p}\vec{p}^\top)\vec{d} \\
&= \sum_{i=1}^{n} p_i d_i^2 - (\vec{d}^\top \vec{p})(\vec{p}^\top \vec{d}) \\
&= \sum_{i=1}^{n} p_i d_i^2 - \left(\sum_{i=1}^{n} p_i d_i \right)^2
\end{aligned}$$

3. Applying Jensen's Inequality
Let X be a discrete random variable that takes values d_i with probability p_i.

$$\sum_{i=1}^{n} p_i d_i = \mathbb{E}[X] \quad \text{and} \quad \sum_{i=1}^{n} p_i d_i^2 = \mathbb{E}[X^2]$$

Consider the function $\phi(t) = t^2$. Since $\phi''(t) = 2 \geq 0$, ϕ is a convex function. Jensen's Inequality states that for a convex function ϕ:

$$\mathbb{E}[\phi(X)] \geq \phi(\mathbb{E}[X])$$

Substituting $\phi(t) = t^2$:

$$\mathbb{E}[X^2] \geq (\mathbb{E}[X])^2$$

Translating this back to our summation form:

$$\sum_{i=1}^{n} p_i d_i^2 \geq \left(\sum_{i=1}^{n} p_i d_i \right)^2 \implies \sum_{i=1}^{n} p_i d_i^2 - \left(\sum_{i=1}^{n} p_i d_i \right)^2 \geq 0$$

Therefore, $\vec{d}^\top H \vec{d} \geq 0$. Since the Hessian is positive semi-definite, $f(\vec{x})$ is convex.

6 Information Theory [35pts]

6.1 Entropy Proofs [20pts]

Here, we'll prove some fundamental ideas in information theory regarding entropy. While you're encouraged to review the slides (we may even spoil some proofs), you should refrain from using any identities you find that involve information theory. In this question, you are expected to derive proofs using only standard rules of probability (and algebra of course), so as to avoid a circular argument, where the identity you apply is simply a corollary of the thing you're trying to prove. To be safe, stick to the following tools:

- **Probability Axioms:** $0 \leq P(X = x) \leq 1$, $\sum_x P(X = x) = 1$
- **Definition of Conditional Probability:** $P(X = x|Y = y) \cdot P(Y = y) = P(X = x, Y = y)$
- **Law of Total Probability:** $P(X = x) = \sum_y P(X = x, Y = y) = \sum_y P(X = x|Y = y) \cdot P(Y = y)$
- **continuity extension of $x \log x$:** $0 \cdot \log_2(0) = 0$
- Any log properties.
- The definitions given below.

Let X be a discrete random variable with probability mass function $P(X = x)$ for $x \in \mathcal{X}$. Recall that *entropy* is defined as:

$$H(X) = -\sum_{x \in \mathcal{X}} P(X = x) \log_2 P(X = x)$$

For two discrete random variables X and Y, recall that *conditional entropy* is defined as:

$$H(X|Y) = \sum_{y \in \mathcal{Y}} P(Y = y) \left[-\sum_{x \in \mathcal{X}} P(X = x|Y = y) \log_2 P(X = x|Y = y) \right]$$

Using these definitions,

a. Prove that $H(X) \geq 0$. [5pts]
By the axioms of probability, for any $x \in X$, $0 \leq P(x) \leq 1$. The logarithm function $\log_2(z)$ is non-positive for $z \in (0, 1]$. Thus:

$$\log_2 P(x) \leq 0$$
$$P(x) \log_2 P(x) \leq 0$$
$$-P(x) \log_2 P(x) \geq 0$$

Since $H(X) = \sum_x -P(x) \log_2 P(x)$ is a sum of non-negative terms, the result must be non-negative:

$$H(X) \geq 0$$

b. Using that result, prove that if $H(X) = 0$, then that random variable X must be *deterministic*[4]. [5pts]
From (a), we know that every term $-P(x) \log_2 P(x)$ is ≥ 0.If the sum $H(X) = 0$, then every individual term in the sum must be 0. the equation $-p \log_2 p = 0$ has solutions only at $p = 0$ and $p = 1$.Therefore, for all x, $P(x) \in \{0, 1\}$. Since $\sum_x P(x) = 1$, it is impossible for all probabilities to be 0. Thus, there must be exactly one outcome x_0 such that $P(x_0) = 1$, and all other outcomes have probability 0. A random variable that takes on a single value with probability 1 is the definition of a deterministic variable.

c. Prove that $H(X) - H(X|Y) = \sum_{x,y} P(X = x, Y = y) \cdot \left(-\log_2 \left(\frac{P(X=x) \cdot P(Y=y)}{P(X=x, Y=y)} \right) \right)$. Start from the given definitions and work algebraically to the result. [5pts]
First, expand $H(X)$ using the Law of Tota Probability $P(x) = \sum_y P(x, y)$:

$$H(X) = -\sum_x P(x) \log_2 P(x) = -\sum_x \left(\sum_y P(x, y) \right) \log_2 P(x) = -\sum_{x,y} P(x, y) \log_2 P(x)$$

Next, expand $H(X|Y)$:

$$H(X|Y) = -\sum_y P(y) \sum_x P(x|y) \log_2 P(x|y) = -\sum_{x,y} P(x, y) \log_2 P(x|y)$$

[4] A random variable X is "deterministic" if and only if there exists some x_0 in its support such that $P(X = x_0) = 1$.

Now, take the difference:

$$\begin{aligned}
H(X) - H(X|Y) &= \left(-\sum_{x,y} P(x,y)\log_2 P(x)\right) - \left(-\sum_{x,y} P(x,y)\log_2 P(x|y)\right) \\
&= \sum_{x,y} P(x,y)\left(\log_2 P(x|y) - \log_2 P(x)\right) \\
&= \sum_{x,y} P(x,y)\log_2\left(\frac{P(x|y)}{P(x)}\right)
\end{aligned}$$

Substitute $P(x|y) = \frac{P(x,y)}{P(y)}$:

$$\begin{aligned}
&= \sum_{x,y} P(x,y)\log_2\left(\frac{P(x,y)}{P(x)P(y)}\right) \\
&= \sum_{x,y} P(x,y)\left(-\log_2\left(\frac{P(x)P(y)}{P(x,y)}\right)\right)
\end{aligned}$$

d. Using that result, <u>prove</u> that $H(X|Y) \leq H(X)$. [5pts]

Hint

Jensen's inequality may be useful for (d). Refer to the previous section for clarity on Jensen's.

Consider the function $f(t) = -\log_2(t)$. Since $f''(t) = \frac{1}{t^2 \ln 2} > 0$, $f(t)$ is strictly convex. Let Z be a random variable taking the value $\frac{P(x)P(y)}{P(x,y)}$ with probability mass $P(x,y)$. The expression from part (c) is the expected value $\mathbb{E}[f(Z)]$. By jensen's inequality for a convex function f: $\mathbb{E}[f(Z)] \geq f(\mathbb{E}[Z])$.

$$\mathbb{E}[Z] = \sum_{x,y} P(x,y)\left(\frac{P(x)P(y)}{P(x,y)}\right) = \sum_{x,y} P(x)P(y) = \left(\sum_x P(x)\right)\left(\sum_y P(y)\right) = 1 \cdot 1 = 1$$

Calculate the RHS $f(\mathbb{E}[Z])$:

$$f(1) = -\log_2(1) = 0$$

Thus, $H(X) - H(X|Y) \geq 0$, which implies $H(X|Y) \leq H(X)$.

6.2 KL Divergence [5pts]

Recall the KL divergence formula:

$$D_{\mathrm{KL}}(P\|Q) = \sum_{x\in\mathcal{X}} P(x)\log_2 \frac{P(x)}{Q(x)}$$

This measure is useful in ML because it almost acts as a distance measure between 2 probability distributions over $\mathcal{X}$, here P and Q. We often want to compare a proposal distribution to a ground-truth distribution (e.g., GANs, classification), and this is a potentially useful tool.

Suppose we have a very fuzzy multi-class classification problem where not even human labelling is 100% conclusive, and we actually want our models to model that uncertainty. We have a sample image and polling from a group of human labellers has given us the following distribution over the set of 4 possible labels:

- True distribution: $P = [0.6, 0.15, 0.2, 0.05]$

3 different models have attempted to predict the consensus with the following outputs:

- Model A predictions: $Q_A = [0.7, 0.2, 0.05, 0.05]$
- Model B predictions: $Q_B = [0.55, 0.1, 0.1, 0.25]$
- Model C predictions: $Q_C = [0.58, 0.2, 0.18, 0.04]$

a. Calculate $D_{KL}(P\|Q_A)$, $D_{KL}(P\|Q_B)$, and $D_{\mathrm{KL}}(P\|Q_C)$. Show your work. [3pts]

1. Model A (Q_A):

$$\begin{aligned}
D_{KL}(P||Q_A) &= 0.6\log_2\left(\frac{0.6}{0.7}\right) + 0.15\log_2\left(\frac{0.15}{0.2}\right) + 0.2\log_2\left(\frac{0.2}{0.05}\right) + 0.05\log_2\left(\frac{0.05}{0.05}\right)\\
&= 0.6\log_2(0.857) + 0.15\log_2(0.75) + 0.2\log_2(4) + 0\\
&\approx 0.6(-0.2224) + 0.15(-0.4150) + 0.2(2)\\
&\approx -0.1334 - 0.0623 + 0.4\\
&\approx 0.2043
\end{aligned}$$

2. Model B (Q_B):

$$\begin{aligned}
D_{KL}(P||Q_B) &= 0.6\log_2\left(\frac{0.6}{0.55}\right) + 0.15\log_2\left(\frac{0.15}{0.1}\right) + 0.2\log_2\left(\frac{0.2}{0.1}\right) + 0.05\log_2\left(\frac{0.05}{0.25}\right)\\
&= 0.6\log_2(1.091) + 0.15\log_2(1.5) + 0.2(1) + 0.05\log_2(0.2)\\
&\approx 0.6(0.1255) + 0.15(0.5850) + 0.2 + 0.05(-2.3219)\\
&\approx 0.0753 + 0.0878 + 0.2 - 0.1161\\
&\approx 0.2470
\end{aligned}$$

3. Model C (Q_C):

$$\begin{aligned}
D_{KL}(P||Q_C) &= 0.6\log_2\left(\frac{0.6}{0.58}\right) + 0.15\log_2\left(\frac{0.15}{0.2}\right) + 0.2\log_2\left(\frac{0.2}{0.18}\right) + 0.05\log_2\left(\frac{0.05}{0.04}\right)\\
&= 0.6\log_2(1.034) + 0.15\log_2(0.75) + 0.2\log_2(1.111) + 0.05\log_2(1.25)\\
&\approx 0.6(0.0489) + 0.15(-0.4150) + 0.2(0.1520) + 0.05(0.3219)\\
&\approx 0.0293 - 0.0623 + 0.0304 + 0.0161\\
&\approx 0.0135
\end{aligned}$$

b. Which model's predictions are "closest" to the true distribution? [2pts]
C

6.3 Entropy and Informativity [10pts]

A streaming service wants to understand user engagement patterns, so they collect data on user subscription tiers and viewing behavior. Consider the following subset.

User ID	Subscription Tier (X)	Binge Watched? (Y)	Device Type (Z)
1	Premium	Yes	Mobile
2	Basic	No	TV
3	Premium	Yes	Mobile
4	Standard	No	TV
5	Basic	No	Mobile
6	Premium	Yes	TV
7	Standard	No	Mobile
8	Premium	Yes	TV

a. Calculate $H(Y)$, the entropy of binge watching behavior. Show your work. [3pts]
$Y =$ Yes: 4 users (1, 3, 6, 8) $\implies P(Y = \text{Yes}) = 4/8 = 0.5$.
$Y =$ No: 4 users (2, 4, 5, 7) $\implies P(Y = \text{No}) = 4/8 = 0.5$.

$$H(Y) = -\sum_{y \in \{Yes, No\}} P(y) \log_2 P(y)$$

$$H(Y) = -[0.5 \log_2(0.5) + 0.5 \log_2(0.5)] = -[0.5(-1) + 0.5(-1)] = -[-1] = 1bit$$

b. Compute $H(Y|X)$, the conditional entropy of binge watching given subscription tier. Show your work. [3pts]

Premium ($X = P$): 4 users. $Y = \{Yes, Yes, Yes, Yes\}$. Since all outcomes are "Yes", this is pure. $H(Y|X = P) = 0$.
Basic ($X = B$): 2 users. $Y = \{No, No\}$. Since all outcomes are "No", this is pure. $H(Y|X = B) = 0$.
Standard ($X = S$): 2 users. $Y = \{No, No\}$. Since all outcomes are "No", this is pure. $H(Y|X = S) = 0$.

$$H(Y|X) = P(X = P)H(Y|X = P) + P(X = B)H(Y|X = B) + P(X = S)H(Y|X = S)$$

$$H(Y|X) = \frac{4}{8}(0) + \frac{2}{8}(0) + \frac{2}{8}(0) = 0bit$$

c. Find $I(X;Y)$, the mutual information between subscription tier and binge watching. Show your work. [2pts]
Using the identity $I(X;Y) = H(Y) - H(Y|X)$:

$$I(X;Y) = 1 - 0 = 1bit$$

d. Suppose we want to make a crude demo model using just one accessible feature to predict binge watching behavior. If we're constrained to use only one feature, subscription tier or device type, which would be more informative and why? [2pts] We compare the mutual information of Subscription Tier (X) vs. Device Type (Z).

Mobile ($Z = M$): 4 users. $Y = \{Yes, Yes, No, No\}$. (2 Yes, 2 No). $H(Y|Z = M) = -[0.5 \log_2 0.5 + 0.5 \log_2 0.5] = 1$.
TV ($Z = T$): 4 users. $Y = \{No, No, Yes, Yes\}$. (2 Yes, 2 No). $H(Y|Z = T) = 1$.

$$H(Y|Z) = 0.5(1) + 0.5(1) = 1$$

$$I(Z;Y) = H(Y) - H(Y|Z) = 1 - 1 = 0bit$$

Since $I(X;Y) > I(Z;Y)$ $(1 > 0)$, subscription tier is the more informative feature. also, it perfectly predicts the target variable, whereas device type provides zero information gain.

7 Entropy Bound [4% Bonus for All]

Earlier, we explored the lower bound on entropy. Now, we'll discuss the upper bound.

Let X be a random variable whose support is the set $\{x_1, x_2, \ldots, x_K\}$. The random variable X will take value x_i with probability p_i, i.e., $P(X = x_i) = p_i$ for $1 \leq i \leq K$. Thus, there exists a set of possible probability distributions for X,

$$\mathcal{P} = \left\{ (p_1, p_2, \cdots, p_K) \in \mathbb{R}^K \;\middle|\; \forall i[p_i \geq 0], \text{ and } \sum_{i=1}^{K} p_i = 1 \right\}$$

This is a set of vectors in $\mathbb{R}^K$ that are constrained by a geometric surface (here, a plane). We call such a set of vectors a *simplex*.

Recall that Shannon entropy, H, is defined on probability distributions where, if X takes a probability distribution $q = (q_1, q_2, \cdots, q_K)$ from the simplex $\mathcal{P}$, then the Shannon entropy of X under q is defined as:

$$H(X_q) = \sum_{i=1}^{K} -q_i \cdot \log_2(q_i)$$

The uniform distribution $u = (u_1, u_2, \cdots, u_K)$ is an element of the simplex $\mathcal{P}$ where $u_i = \frac{1}{K}$ for all $1 \leq i \leq K$ (all values have the same probability). Prove that the uniform distribution is an element of this simplex that, when used as a probability distribution for X, maximizes $H(X)$. [4% EC]

Let $p = (p_1, p_2, \ldots, p_K)$ be an arbitrary probability distribution in the simplex $\mathcal{P}$. Let $u = (1/K, 1/K, \ldots, 1/K)$ be the uniform distribution.

1. compute the entropy of the uniform distribution:

$$H(u) = -\sum_{i=1}^{K} \frac{1}{K} \log_2\left(\frac{1}{K}\right) = -\log_2\left(\frac{1}{K}\right) \sum_{i=1}^{K} \frac{1}{K} = \log_2 K$$

2. consider the difference between the entropy of p and the maximum possible value $\log_2 K$:

$$H(p) - \log_2 K = -\sum_{i=1}^{K} p_i \log_2 p_i - \log_2 K$$

Since $\sum p_i = 1$, multiply the $\log_2 K$ term by the sum:

$$\begin{aligned} H(p) - \log_2 K &= -\sum_{i=1}^{K} p_i \log_2 p_i - \sum_{i=1}^{K} p_i \log_2 K \\ &= \sum_{i=1}^{K} p_i(-\log_2 p_i - \log_2 K) \\ &= \sum_{i=1}^{K} p_i \log_2\left(\frac{1}{p_i K}\right) \end{aligned}$$

3. apply Jensen's Inequality. The function $f(t) = \log_2(t)$ is concave. Jensen's inequality states that for a concave function, $\mathbb{E}[f(Y)] \leq f(\mathbb{E}[Y])$. Let Y be a random variable taking on values $y_i = \frac{1}{p_i K}$ with probabilities p_i.

$$\sum_{i=1}^{K} p_i \log_2\left(\frac{1}{p_i K}\right) \leq \log_2\left(\sum_{i=1}^{K} p_i \cdot \frac{1}{p_i K}\right)$$

$$\sum_{i=1}^{K} p_i \cdot \frac{1}{p_i K} = \sum_{i=1}^{K} \frac{1}{K} = K \cdot \frac{1}{K} = 1$$

$$H(p) - \log_2 K \leq \log_2(1) = 0$$

$$H(p) \leq \log_2 K$$

Since $H(p) \leq H(u)$ for all $p \in \mathcal{P}$, the uniform distribution maximizes the Shannon entropy.